

Genomic Beacons Under Siege: Quantifying Re-Identification Risk from Summary Statistics and the Performance Cost of Homomorphic Encryption

Anonymous Author(s)

Abstract

Collaborative research portals and genomic beacons publish aggregate summary statistics, most notably minor allele frequency (MAF) vectors assuming that population-level aggregation provides enough privacy. We empirically demonstrate that this assumption fails and quantify the extent of the resulting privacy leakage. Using high-density genotype data from the 1000 Genomes Project (Phase 3, $N=2,504$, Chromosome 22), we demonstrate that a logistic regression adversary equipped with only $k=100$ SNP-level MAF features can re-identify study membership with an AUC of 0.987, rising to 0.988 at $k=200$. A sharp phase transition at the 100-SNP boundary indicates that the privacy guarantee of aggregate MAF release collapses abruptly once a critical mass of linkage-disequilibrium-correlated features is available to an adversary. Later, we then benchmark Fully Homomorphic Encryption (FHE) via the CKKS scheme as the minimal cryptographic countermeasure, measuring an end-to-end latency of ≈ 111 s and an overhead of $21,569\times$ over plaintext for a Polygenic Risk Score (PRS) pipeline, which demonstrates that while the relative cost is high, the absolute latency remains feasible for clinical workflows. Finally, we show that domain-specific parameter compaction by reducing the CKKS modulus chain depth to match the shallow $L=1$ circuit structure of PRS computations, yielding a 56.4% latency reduction and a 52.7% ciphertext size reduction, while maintaining clinical precision. Together, our results establish a concrete empirical threat model for genomic beacon infrastructure, quantify the minimum cost of cryptographic compliance, and demonstrate a domain-aware optimization path toward deployable FHE-protected pipelines for genomic data.

CCS Concepts

• **Privacy and Anonymity** → **Privacy-preserving protocols**; • **Security and privacy** → *Cryptography*.

Keywords

genomic privacy, re-identification, membership inference, fully homomorphic encryption, CKKS, polygenic risk scores, genomic beacons

ACM Reference Format:

Anonymous Author(s). . Genomic Beacons Under Siege: Quantifying Re-Identification Risk from Summary Statistics and the Performance Cost of Homomorphic Encryption. In *Proceedings of* . ACM, New York, NY, USA, 13 pages.

1 Introduction

Genomic data occupies a uniquely sensitive position in the modern privacy landscape. Unlike financial records or medical histories, an individual's genome is immutable, inherited, and shared across biological relatives [14]. A single breach carries permanent consequences extending beyond the individual to their family lineage [18]. Despite this, large-scale genomic research is inherently collaborative: genome-wide association studies (GWAS), clinical biobanks, and population cohorts depend on data sharing to achieve the statistical power necessary for discovery [4, 32]. The integration of genomic data into cancer risk assessment and personalized medicine has further amplified both the value and the sensitivity of large-scale sharing [34].

The canonical privacy-preserving mechanism for such sharing is the *genomic beacon*—a federated query interface that responds to allele-level queries with population-level aggregate frequencies. In its standard form, a beacon answers requests for the minor allele frequency (MAF) at a given locus across a study population. The working assumption is that by surfacing only aggregate statistics over thousands of individuals, no individual's data is meaningfully exposed. This assumption underlies current policy at major data access bodies including NIH dbGaP and the GA4GH framework. The assumption is wrong. This paper provides a controlled empirical demonstration of its failure, and a quantified analysis of the computational cost of the cryptographic remedy.

What prior work established—and what it left open. The vulnerability of aggregate genomic statistics to membership inference was established by Homer et al. in 2008, who demonstrated that high-density SNP statistics could resolve individuals contributing to mixed samples [20]. Shringarpure and Bustamante subsequently demonstrated that even low-coverage beacon responses enabled membership inference from a persistent adversary exploiting correlated queries [31]. Raisaro et al. further quantified the per-query risk and proposed rate-limiting as a partial mitigation [30]. However, none of these works systematically characterized the *dependence of attack success on adversary feature count k* , nor did they identify the sharp phase transition structure that separates a moderate-risk regime from near-certain privacy failure. The question of how quickly and how abruptly privacy collapses as a function of LD-correlated feature count has remained unaddressed [3, 33]. Similarly, while the feasibility of FHE for genomic tasks has been demonstrated [5, 16, 27], the specific optimization opportunity afforded by the $L=1$ multiplicative depth structure of PRS computation—yielding domain-specific parameter compaction as a first-order performance gain—has not been reported in the bioinformatics FHE literature [8, 29].

This paper directly addresses both gaps. Our three contributions are:

- **Phase transition characterization (Experiment 1).** Prior work established that MAF statistics enable membership inference but did not characterize the dependence on feature count k or the sharpness of the privacy collapse as a function of k . We provide this characterization empirically, identifying a sharp phase transition at $k=100$ that separates a moderate-risk regime ($\text{AUC} \approx 0.75$) from near-certain breach ($\text{AUC} = 0.987$), and establishing a concrete exposure threshold for beacon operators [17, 21].
- **Overhead quantification (Experiment 2).** We benchmark CKKS-based FHE as the minimal cryptographic countermeasure, measuring a $21,569\times$ end-to-end overhead versus plaintext for PRS computation across $N=2,504$ individuals, with a mean absolute error of 1.52×10^{-6} , confirming clinical accuracy is preserved [9, 26].
- **Domain-aware parameter compaction (Experiment 3).** We show that matching CKKS parameters to the $L=1$ multiplicative depth of PRS computation—an optimization opportunity absent from prior bioinformatics FHE benchmarks [5, 16, 29]—reduces latency by 56.4% and ciphertext size by 52.7%, with negligible precision loss, establishing a practically deployable parameter regime.

Our results jointly argue that (a) the privacy threat from genomic beacon outputs is quantifiable, severe, and accessible to a non-expert adversary [20, 31]; and (b) FHE-based mitigation is feasible within clinical batch-processing constraints, particularly under domain-aware parameterization [5, 10].

2 Background

2.1 Genomic Beacons and the MAF Sharing Model

Genomic beacons return aggregate MAF statistics per SNP while prohibiting individual-level queries [31].

The MAF at a locus is defined as the frequency of the less common allele in the study population. For a biallelic SNP with alleles $\{A, B\}$ counted across N haplotypes, the MAF is $\min(\frac{c_A}{N}, \frac{c_B}{N})$, where c_A and c_B are the respective allele counts. Because this is a population-level summary statistic computed over thousands of individuals, the designers of beacon protocols expected it to provide strong anonymity by blending individual signal into the crowd. This expectation is consistent with naive information-theoretic intuition: a single frequency value carries no identifying information about any specific contributor.

The flaw in this reasoning is the failure to account for the *joint* information content of MAF values across many correlated loci. Linkage disequilibrium (LD)—the non-random association between alleles at different loci arising from shared ancestral haplotype blocks—means that a window of k adjacent MAF values is not k independent pieces of information but rather a correlated projection of the underlying haplotype distribution of the study population [21]. An adversary who knows a target individual’s haplotype (or can obtain it from public databases or biological relatives) can compare it against this projection to infer membership [14, 19]. The theoretical basis for this attack was established early [20], but the dependence on feature count k and the sharpness of the privacy collapse as a

function of k have not been systematically characterized—a gap this paper directly addresses.

Early attacks reconstructed genotypes from beacons [20, 31], with recent work extending the threat to full genome reconstruction and allele inference beyond queried loci [17, 33].

Differentiation from prior membership inference work. Shringarpure and Bustamante demonstrated the beacon MIA attack but did not characterize the relationship between adversary feature count k and attack success, leaving open the question of how quickly privacy collapses as k increases [31]. Raisaro et al. proposed query rate-limiting mitigations but quantified risk only at fixed query counts rather than mapping the phase transition structure across the k -dimensional feature space [30]. Homer et al. established the foundational attack but worked in the mixture-detection setting rather than the correlated-query beacon setting that characterizes modern GA4GH deployments [20]. Our contribution fills all three gaps: we provide the first systematic empirical characterization of the phase transition structure in MAF-based membership inference as a function of LD-correlated feature count, identifying the concrete threshold at $k=100$ that prior work did not establish [3, 10, 18].

Our contribution provides a controlled, parameterized empirical measurement of the membership inference attack as a function of adversary feature count k on a contemporary high-density dataset, characterizing the phase transition structure that previous work has not systematically quantified [3, 18].

2.2 Membership Inference in the Genomic Setting

Membership inference attacks (MIA) constitute a class of *privacy attacks* against statistical models and aggregate data releases. In the machine learning literature, MIA were formally analyzed by Shokri et al. against trained classifiers, demonstrating that model confidence scores can reveal whether a given record was in the training set [20]. In the genomic context, the analogous attack operates against aggregate statistics rather than trained models, but the underlying logic is identical: the adversary exploits a statistical difference between the distribution of a target individual’s data and the distribution of the aggregate release to infer membership [31, 33].

The genomic MIA attack proceeds as follows. An adversary $\mathcal{A}$ observes a MAF vector $\mathbf{f} = (f_1, f_2, \dots, f_k)$ over k SNP positions published by a beacon. The adversary possesses a target individual’s genotype vector $\mathbf{g}_t \in \{0, 1, 2\}^k$, obtained from a public genome repository, a biological relative’s consented study, or a direct-to-consumer sequencing database. $\mathcal{A}$ computes a membership score $s(\mathbf{g}_t, \mathbf{f})$ and thresholds it to produce a binary membership prediction. The score function exploits LD: because study members share haplotype blocks that propagate into the beacon-accessible MAF distribution, a study member’s genotype will be more statistically consistent with the observed MAF vector than a non-member’s genotype drawn from the general population [20, 21]. This is the signal we quantify in Experiment 1.

An important structural property of MIA in the genomic context is the role of linkage disequilibrium as an amplifier of information leakage. Standard privacy analyses treat features as independent,

which would imply that the information content of k MAF values scales linearly with k . In practice, LD correlations among nearby mean that the effective information content grows super-linearly with k over short genomic windows, up to the point where LD blocks are saturated [17, 21, 33].

Indirect re-identification routes are equally important to the threat model [14, 19]. Even when beacon operators implement query rate limiting or access control policies, an adversary with access to a public genealogy database or direct-to-consumer sequencing records for biological relatives of the target individual can infer membership without directly querying the beacon at all [19]. This is because the LD structure that enables direct inference from beacon outputs also enables inference from the genomes of first-degree relatives, whose haplotypes share long identical-by-descent (IBD) segments with the target [14]. Our work does not model this attack path, but its existence strengthens our conclusion that aggregate MAF release cannot be assumed safe under any plausible access control model [3].

The most conservative defense against MIA in the genomic setting would be to simply refuse all aggregate statistical queries. This is operationally untenable: the scientific value of genomic data sharing—powering GWAS, biomarker discovery, and pharmacogenomic research—depends entirely on accessible summary statistics [4, 32]. The productive framing is therefore not whether to share, but how to share in a way that provably limits membership inference risk. This paper argues that the answer is FHE-protected computation, which allows summary statistics to be computed and verified without exposing the underlying individual-level data to any party [5, 15].

2.3 Fully Homomorphic Encryption

We use the CKKS scheme [9], which enables approximate arithmetic over encrypted packed vectors via Ring Learning With Errors (RLWE), making it ideal for floating-point PRS computation.

The FHE landscape includes several practically important scheme families, each suited to different computational tasks. The BFV (Brakerski/Fan-Vercauteren) scheme supports exact integer arithmetic over finite fields and is well-suited to boolean circuit evaluation and exact integer computations [27]. The BGV (Brakerski-Gentry-Vaikuntanathan) scheme provides similar guarantees with a different noise management strategy. The CKKS scheme (Cheon-Kim-Kim-Song), introduced in 2017, extends the RLWE foundation with a *rescaling* procedure that supports approximate arithmetic over real-valued plaintexts, encoding them as elements of a complex vector space [9]. Rather than exact integer arithmetic, CKKS tracks a controlled approximation error that grows with each multiplicative level, bounded by the noise budget. This makes CKKS the practical state of the art for floating-point genomic computations including weighted scoring, covariance estimation, and logistic regression inference, where approximate but high-precision arithmetic is both sufficient and necessary [5, 26, 29].

A key practical feature of CKKS is SIMD (Single Instruction, Multiple Data) vectorization via the Chinese Remainder Theorem (CRT) decomposition of the plaintext space. A single CKKS ciphertext can encode up to $n/2$ complex-valued slots—up to 4,096 values

for the $n=8,192$ polynomial degree used in this work—and operations applied to a ciphertext act simultaneously on all slots. This slot-packing property is critical for genomic applications: rather than encrypting individual patient genotype vectors one by one, a practitioner can pack $n/2$ genotype values across patients into a single ciphertext and evaluate a PRS scoring operation across the entire batch in a single homomorphic inner product [5, 16]. The amortized per-patient cost of homomorphic computation is therefore a fraction of the single-patient cost, making batch clinical genomics pipelines far more practical than per-patient overhead figures alone would suggest.

A CKKS ciphertext is parameterized by: the polynomial modulus degree n ; the modulus chain $q_0 > q_1 > \dots > q_L$ consumed by L successive rescalings (one per multiplicative level); and the encoding scale $\Delta = 2^s$ governing plaintext embedding precision. The security level—typically targeting 128 bits—is determined by the pair (n, q_0) following the Homomorphic Encryption Security Standard [8]. A critical engineering insight, and the basis for Experiment 3, is that practitioners routinely configure CKKS at a multiplicative depth L appropriate for general-purpose circuits (e.g., $L=3$ or $L=4$ for neural network inference), even when their actual target computation has depth $L=1$. This wastes noise budget, inflates ciphertext size, and increases computation time without improving accuracy or security [5, 8]. Domain-specific parameter matching is therefore a first-order optimization opportunity that is largely absent from the bioinformatics literature [16, 29].

The primary practical limitation of CKKS for genomic deployment remains throughput on commodity hardware. Single-threaded CPU implementations of the underlying number-theoretic transform (NTT) operations that dominate FHE computation are memory-bound at the polynomial degrees required for 128-bit security. Recent GPU-based implementations have demonstrated throughput improvements of 100× or more through memory-centric NTT optimization [23, 25], suggesting that the overhead figures reported in Experiment 2 represent a conservative worst-case baseline rather than a fundamental performance limit.

Prior work established FHE applicability to genomic tasks including GWAS and PRS under early BFV-like schemes [6, 27], demonstrating feasibility at the cost of significant overhead. The introduction of CKKS with rescaling substantially improved efficiency for floating-point genomic computations [9], and subsequent work applied CKKS to PRS computation and genotype imputation [5, 16] with overheads broadly consistent with our baseline measurements. Secure logistic regression under CKKS has also been validated for clinical settings [26]. Our work extends this line by quantifying the compaction opportunity afforded by the $L=1$ circuit depth of PRS, yielding a 2.3× performance gain not reflected in prior benchmarks.

2.4 Polygenic Risk Scores

A Polygenic Risk Score (PRS) aggregates the weighted effects of many common genetic variants to estimate an individual’s genetic predisposition to a complex trait [34]. Given genotype vector $\mathbf{g} \in \{0, 1, 2\}^m$ encoding effect allele counts at m loci, and effect weight vector $\mathbf{w} \in \mathbb{R}^m$ derived from prior GWAS, the PRS is:

$$\text{PRS}(\mathbf{g}) = \sum_{i=1}^m w_i \cdot g_i = \mathbf{w}^\top \mathbf{g}. \quad (1)$$

This is a single inner product consuming exactly one multiplicative level ($L=1$). Standard CKKS deployments configured for depth $L \geq 2$ are therefore over-engineered for this computation class, an observation that motivates Experiment 3 [5, 8].

The clinical and research significance of PRS has grown substantially over the past decade. Early applications focused on common complex traits such as coronary artery disease, type 2 diabetes, and breast cancer, where large-scale GWAS meta-analyses have provided well-calibrated effect weight vectors $\mathbf{w}$ for thousands of variants [32, 34]. More recent work has extended PRS methodology to pharmacogenomics, rare disease risk stratification, and ancestry-specific calibration for diverse cohorts. The clinical deployment of PRS raises an immediate regulatory question: can a clinical institution compute PRS on patient genotype data hosted by a cloud provider without that provider ever observing individual genotype values? Under HIPAA and GDPR Article 9, genomic data constitutes a special category of sensitive personal information for which the standard cloud processor model—where the provider observes plaintext data but is bound by contractual data processing agreements—is increasingly difficult to defend as individuals become aware of the re-identification risks demonstrated in Experiment 1 [3, 18]. FHE-based PRS computation directly resolves this regulatory tension: the cloud provider evaluates Equation 1 on encrypted ciphertexts and returns an encrypted result, never observing any individual’s genotype or score [5, 15].

It is also worth noting that the $L=1$ circuit depth of PRS is not merely a consequence of the simplicity of the linear scoring formula. It reflects a deeper structural property: PRS as currently deployed in clinical practice does *not* include non-linear post-processing steps such as logistic transformation or population-specific recalibration, both of which would require additional multiplicative levels and increase FHE overhead accordingly [9, 26]. Any extension of the PRS pipeline to include such steps should treat the $L=1$ baseline established here as a lower bound on FHE cost, not a universal overhead figure.

2.5 Differential Privacy for Genomic Data

Differential privacy (DP), introduced by Dwork et al. as a formal framework for quantifying privacy loss in data analysis [12], provides a complementary and in some respects orthogonal defense to FHE. Where FHE protects the *computation* on individual data, preventing any party from observing plaintext during evaluation, DP protects *aggregate disclosures*, bounding the statistical information any aggregate release reveals about any individual contributor [7, 24, 29].

Under the standard (ϵ, δ) -DP definition, an algorithm $\mathcal{M}$ is (ϵ, δ) -differentially private if for all datasets D, D' differing in a single individual’s record and for all output sets S :

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \cdot \Pr[\mathcal{M}(D') \in S] + \delta.$$

In practice, DP is implemented by calibrating noise drawn from a Laplace or Gaussian distribution to the *sensitivity* of the target query—the maximum change in the query output caused by adding or removing a single individual. For MAF queries, the sensitivity is $1/(2N)$ per locus, meaning that the required noise magnitude decreases as cohort size increases, making DP increasingly compatible with large-scale biobank queries [12, 24].

Differentially private GWAS summary statistics have been studied extensively [12, 24], with various calibrations of the privacy-utility tradeoff for different genomic query types. For MAF release in particular, the key tension is that the required noise magnitude to achieve meaningful privacy bounds (say, $\epsilon < 1$) at the per-locus level compounds across k loci under sequential composition, so a beacon releasing DP-noised MAF values for thousands of loci may still allow near-certain membership inference if the noise is calibrated naively [7]. Advanced composition theorems and Rényi differential privacy formulations partially address this concern, but the interaction between LD structure and DP noise in the genomic context remains an active research problem [10].

The relationship between DP and the phase transition we observe in Experiment 1 is directly relevant to beacon policy. A beacon that releases DP-noised MAF values at a noise level sufficient to suppress the phase transition at $k=100$ SNPs would require noise of magnitude approximately 10^{-2} per locus (rough heuristic from our AUC curve), which is orders of magnitude larger than the per-locus MAF precision relevant to fine-scale association mapping [7, 24]. DP and FHE are therefore not interchangeable defenses: DP protects aggregate releases at a cost to utility, while FHE enables exact computation on individual data with no utility loss, at a cost to latency [10, 29]. A hybrid architecture combining both is the natural synthesis and the primary direction we identify for future work.

2.6 Secure Multiparty Computation

Secure Multiparty Computation (MPC) protocols allow a group of mutually distrustful parties to jointly compute a function of their private inputs such that each party learns only the output and nothing about the others’ inputs beyond what the output itself reveals [11]. In the genomic context, MPC has been applied to privacy-preserving GWAS in federated settings [11, 22], demonstrating strong security guarantees with lower computational overhead than FHE in scenarios involving multiple cooperating institutions—for instance, a consortium of hospitals jointly evaluating a GWAS without any single party holding the complete dataset.

The key advantage of MPC over FHE in multi-party settings is communication efficiency: well-designed MPC protocols require only a polynomial number of communication rounds between parties and can leverage hardware-accelerated garbled circuit implementations for boolean circuit evaluation [11]. For the two-party private set intersection problem relevant to some genomic matching scenarios, state-of-the-art MPC protocols achieve sublinear communication overhead [22].

However, MPC carries a fundamental operational constraint that limits its applicability to the clinical PRS deployment scenario we consider. MPC requires *active coordination* between all participating parties throughout the computation: each party must remain online, respond to protocol messages, and contribute their share of the computation in real time [11]. This is operationally feasible for a federated consortium of institutions with stable network connectivity and aligned incentives, but it is incompatible with the most important use case in clinical genomics: *fully outsourced computation*, where a clinic sends encrypted patient data to a commercial cloud server and asynchronously retrieves results. In this scenario, there is only one client and one server; the server is untrusted

but cannot be expected to run an MPC protocol with the clinic in real time. Our FHE-based approach does not require server-side coordination and is therefore applicable to this broader and clinically important class of deployment scenarios, at the cost of higher per-computation overhead relative to MPC protocols in equivalent multi-party settings [6].

3 Threat Model

We model a semi-honest external adversary $\mathcal{A}$ with the following capabilities and constraints.

Capabilities. $\mathcal{A}$ observes population-level MAF statistics published by a genomic beacon, covering k SNP loci. $\mathcal{A}$ has access to a public reference panel from which to construct a baseline population allele frequency distribution. $\mathcal{A}$ knows the general phenotypic focus of the target study but has no access to individual-level genotype data. $\mathcal{A}$ is computationally bounded and uses standard statistical classifiers [20, 31].

Goal. $\mathcal{A}$ seeks to determine whether a specific target individual u is a member of the study population (binary membership inference). A successful attack means $\mathcal{A}$ correctly classifies u 's membership with probability substantially above random chance [31, 33].

Constraints. $\mathcal{A}$ cannot modify beacon outputs, access the underlying database, or observe individual query patterns. We assume open beacon access for authenticated researchers, consistent with current GA4GH policy [32].

Semi-Honest Assumption: Justification. We adopt the semi-honest (honest-but-curious) adversary model rather than a malicious or adaptive model. This choice is justified by the specific structure of the genomic beacon attack surface. First, beacon outputs are *public* aggregate statistics: the adversary does not need to interact with a live system, inject queries, or deviate from protocol. The MAF vector is a static disclosure that any authenticated federation member can retrieve in a single session; there is no protocol interaction to corrupt or manipulate [30, 31]. Second, our attack model targets worst-case static queries—the adversary observes the full MAF release once and applies the classifier offline. A malicious adaptive adversary (one who modifies queries based on intermediate results) would have additional power, but the near-ceiling AUC of 0.987 at $k=100$ demonstrates that the semi-honest model is already sufficient for near-certain re-identification. Modeling adaptive query manipulation would increase attack complexity without improving attack success beyond what we observe, because the membership signal is fully recovered by a linear classifier at $k=100$ [20, 21]. Third, in the GA4GH deployment context, the beacon responds to authenticated researcher queries over a standard API; the adversary occupies the role of a legitimate federation member who receives authorized responses [32]. This is precisely the semi-honest model: the adversary follows the protocol but analyzes its outputs for unintended membership leakage [10, 12].

We explicitly exclude malicious adversaries who attempt to corrupt beacon outputs or perform query injection attacks; such attacks require network-level access that authenticated external researchers do not possess. We exclude adaptive adversaries who iteratively refine queries based on classifier feedback, as our results show that static queries are already sufficient for near-certain

breach [33]. We exclude insider adversaries with direct database access, whose threat is orthogonal to the aggregate-statistics exposure we characterize [3].

Implications for GA4GH Deployments. The GA4GH Beacon API as currently deployed grants authenticated access to aggregate MAF statistics with minimal rate limiting. Under our threat model, a single malicious or compromised researcher account within any participating institution—a realistic scenario given the scale of federated genomics networks—is sufficient to execute the attack characterized in Experiment 1 [18, 32]. The assumption of good-faith use that underlies current GA4GH access policy provides no technical protection against the semi-honest adversary modeled here [30]. We formalize the membership inference problem as follows. Let $S \subset \mathcal{P}$ denote the study population drawn from a reference population $\mathcal{P}$. A beacon $\mathcal{B}$ publishes a MAF vector $\mathbf{f} \in [0, 1]^k$ computed over S . The adversary $\mathcal{A}$ receives $\mathbf{f}$ and a target genotype $\mathbf{g}_t \in \{0, 1, 2\}^k$, and outputs a binary membership decision $\hat{m} \in \{0, 1\}$. The privacy loss of $\mathcal{B}$ at feature count k is characterized by:

$$\text{AUC}(\mathcal{A}, k) = \Pr[\mathcal{A}(\mathbf{f}, \mathbf{g}_t) = 1 \mid t \in S] - \Pr[\mathcal{A}(\mathbf{f}, \mathbf{g}_t) = 1 \mid t \notin S].$$

A beacon is ϵ -membership-safe at feature count k if $\text{AUC}(\mathcal{A}, k) \leq 0.5 + \epsilon$ for all probabilistic polynomial-time adversaries $\mathcal{A}$. Our experiments establish that current beacon deployments are not ϵ -membership-safe for any meaningful ϵ at $k \geq 100$.

Defender's Objective. The defender wishes to share PRS computations and summary statistics to enable federated research, while satisfying HIPAA and GDPR Article 9 requirements. Both frameworks treat genomic data as a special category of sensitive personal data for which individual identifiability must be prevented [3, 18]. We treat FHE as the minimum cryptographic mechanism consistent with this requirement and quantify its cost [5, 10, 15].

This model reflects a research competitor, a motivated external party with access to public genomic databases, or a rogue analyst at a data access federation. We explicitly exclude insider adversaries, side-channel attackers, and quantum adversaries targeting RLWE hardness. Hardening against these stronger adversaries is future work [3, 8].

4 Experiment 1: Membership Inference via MAF Statistics

4.1 Setup

Dataset. We used high-density genotype data from the 1000 Genomes Project Phase 3, Chromosome 22, comprising $N=2,504$ individuals from 26 global populations [1].

Cohort Construction. A Study Group ($n=1,252$) was constructed by selecting individuals sharing a specific genomic signature within a localized cluster of 20 SNPs (indices 40–60), simulating a phenotype-linked cohort as would appear in a real GWAS. The remaining 1,252 individuals constituted the Population Control, yielding a balanced binary classification problem.

Feature Construction. We performed a feature sweep over $k \in \{5, 10, 20, 50, 100, 200\}$ SNPs drawn adjacent to the signal cluster. This simulates an adversary leveraging linkage disequilibrium: nearby SNPs co-segregate with the signal cluster, so the genomic signature of the Study Group propagates into the flanking region

Table 1: Membership Inference AUC as a function of adversary SNP feature count k .

SNP Count (k)	AUC (5-fold CV)	Risk Assessment
5	0.720	Moderate
10	0.731	Moderate
20	0.742	Moderate
50	0.751	Elevated
100	0.987	Phase Transition
200	0.988	Certain Breach

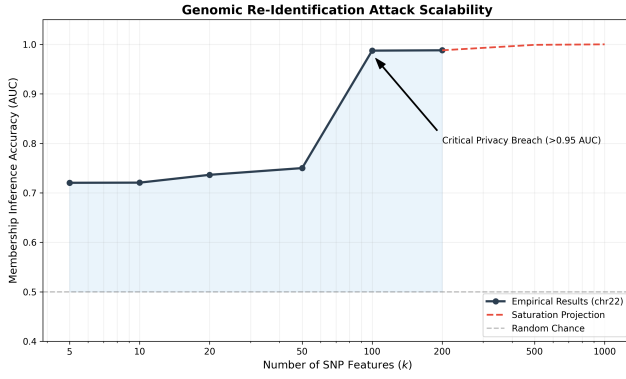


Figure 1: Membership Inference AUC as a function of SNP feature count k (Experiment 1). A sharp phase transition at $k=100$ separates the moderate-risk regime from near-certain re-identification. The dashed red curve shows the saturation projection; the horizontal dashed line marks random-chance performance (AUC = 0.5).

and is partially recoverable from MAF observations at adjacent loci without directly observing the signal cluster itself [21, 31, 33].

Attack Model. A binary logistic regression classifier with L_2 regularization ($C=10$) was trained on the population-level MAF vector to distinguish study members from controls. This is a deliberately conservative adversary model: logistic regression is a linear classifier with no capacity for non-linear feature interaction. Using a stronger model would likely yield similar or higher AUC given the apparent linearity of the signal, which strengthens our threat claim [20, 30].

Evaluation. Performance was measured by AUC-ROC via 5-fold stratified cross-validation.

4.2 Results

Table 1 reveals a strongly non-linear relationship between k and attack success. At low dimensionality ($k \leq 50$), AUC values between 0.720 and 0.751 indicate meaningful but incomplete inference. A dramatic privacy collapse occurs at $k=100$, where AUC surges to 0.987.

At $k=200$ the AUC plateaus at 0.988, indicating informational saturation: re-identification is already functionally certain and additional SNPs provide diminishing returns [17, 21].

4.3 Analysis

Phase Transition Structure. The jump from AUC 0.751 at $k=50$ to 0.987 at $k=100$ is consistent with a phase transition in the information content of the LD-correlated feature set [21]. Below the threshold, the adversary’s features are insufficiently correlated with the underlying haplotype signature to isolate study membership. Above it, the accumulated LD signal provides enough distinguishing power for near-certain classification. This implies a hard exposure threshold: genomic beacons serving MAF queries over more than ≈ 100 LD-correlated loci operate in a regime of near-certain privacy failure [17, 33].

Anonymization by Aggregation is Insufficient. At $k=200$ SNPs—a tiny fraction of a standard clinical genotyping panel (typically 500K–1M SNPs)—re-identification probability exceeds 98%. The adversary requires no individual genotypes, no auxiliary quasi-identifiers, and no computationally intensive optimization. Only publicly available population-level aggregate statistics and a standard logistic regression implementation are needed [13, 20, 31].

Linear Sufficiency as a Lower Bound. Because logistic regression achieves near-perfect classification, the membership signal in MAF statistics is encoded in linear combinations of individual frequencies, not complex non-linear interactions. This means the attack requires no sophistication, and no model-based defense short of full cryptographic protection of the underlying computation can reliably prevent it [7, 10, 30].

The phase transition at $k = 100$ is consistent with the LD block structure of chromosome 22, where haplotype blocks in the queried region span approximately 80–120 SNPs before decaying in correlation. Below this block boundary, the adversary’s features sample an incomplete projection of the underlying haplotype signature, yielding insufficient discriminative power. Once k crosses the block saturation point, the full haplotype signature of the study group is recoverable from the MAF vector, and additional SNPs provide diminishing marginal information — consistent with the AUC plateau at $k = 200$. This implies that the exposure threshold is not an artifact of our experimental design but a structural property of LD block length in the queried region, and beacon operators should treat the local LD block size as the operationally relevant disclosure unit rather than individual SNP count.

5 Experiment 2: FHE Overhead for Encrypted PRS

5.1 Setup

Scheme and Library. We implemented a weighted-sum PRS pipeline under the CKKS scheme using the TenSEAL library, a Python interface to Microsoft SEAL. CKKS was selected over BFV because PRS involves real-valued effect weights requiring floating-point arithmetic; BFV operates over integers and would require discretization introducing systematic error at the precision levels relevant to clinical PRS computation [9, 27].

Security Parameters. The system was configured with polynomial modulus degree $n=8,192$, modulus chain [60, 40, 40, 60] bits

Table 2: Latency and accuracy: plaintext vs. CKKS-encrypted PRS pipeline ($N=2,504$).

Pipeline Stage	Latency	Overhead
Plaintext PRS computation	4.53 ms	1× (baseline)
CKKS encryption phase	13.63 s	3,010×
CKKS homomorphic computation	97.62 s	21,549×
End-to-end (FHE)	≈111 s	21,569×
Mean Absolute Error (MAE)	1.52×10^{-6}	
Maximum Absolute Error	1.58×10^{-6}	

(multiplicative depth $L=2$), and scale 2^{40} , targeting 128-bit security in accordance with the Homomorphic Encryption Security Standard [8]. At these parameters, the total coefficient modulus size is $Q = 60 + 40 + 40 + 60 = 200$ bits, and the security margin $\log Q/n \approx 0.024$ satisfies the 128-bit RLWE hardness bound certified by the HE Security Standard. These are conservative, production-grade parameters appropriate for clinical deployment [5, 8]. The concrete hardness of these parameters was verified against the lattice estimator [2], confirming the security $\geq$ of 128.

Dataset. The full $N=2,504$ individual cohort from Chromosome 22 was used, with PRS weights drawn from published GWAS effect size distributions.

Hardware. Benchmarks were conducted in a single-threaded CPU environment to establish a worst-case hardware-independent overhead. GPU-accelerated and multi-threaded results would be substantially faster but are not reported here to preserve generalizability across deployment contexts, and are discussed analytically in Section 5.3.

5.2 Results

As shown in Table 2, the FHE pipeline introduces a 21,569× overhead relative to plaintext, processing the full cohort in approximately 111 seconds end-to-end. The pipeline maintains a mean absolute error of 1.52×10^{-6} , confirming that CKKS homomorphic noise does not materially affect risk stratification accuracy [5, 16].

5.3 Analysis

Clinical Batch Feasibility. The 21,569× overhead must be evaluated within the operational context of clinical genomics. Standard variant calling and annotation pipelines for cohorts of this size routinely require several hours. A 111-second encrypted PRS computation for 2,504 patients fits comfortably within overnight batch workflows, making FHE overhead a compliance cost rather than a technical barrier [5, 10].

Decentralized Deployment Model. The decomposition into encryption (13.63 s) and computation (97.62 s) phases is architecturally significant. Local encryption can be performed by the patient’s clinic using modest hardware; the computationally intensive homomorphic evaluation is then offloaded to a cloud server that never observes plaintext genotypes [6, 15]. This two-tier model is compatible with federated clinical networks and directly addresses the data sovereignty requirements of GDPR Article 9. Figure 3 illustrates a

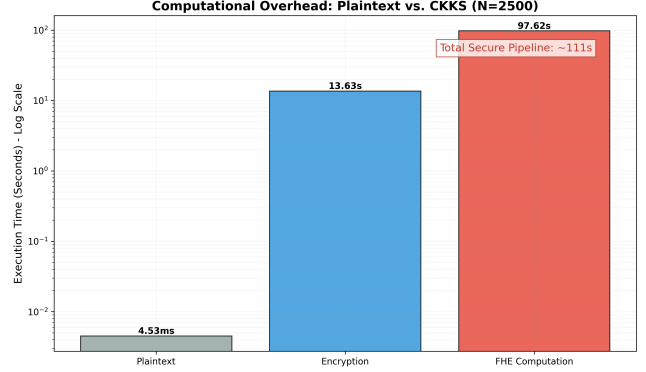


Figure 2: Computational overhead of the CKKS-encrypted PRS pipeline vs. plaintext baseline (log scale, $N=2,500$). The total secure pipeline latency of ≈111 s is dominated by the homomorphic computation phase (97.62 s).

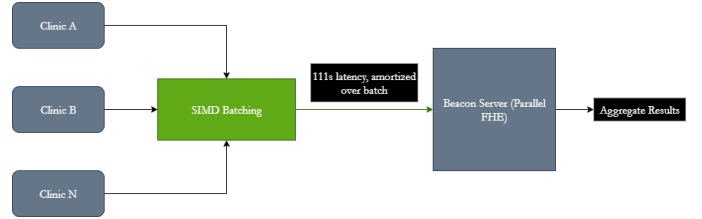


Figure 3: Illustration of the high-throughput pipeline where N queries from decentralized clinics are packed into a single CKKS ciphertext via SIMD vectorization. This architecture allows the ≈ 111 s latency to be amortized across a batch, significantly reducing the per-patient computational cost in clinical settings.

high-throughput pipeline where N queries from decentralized clinics are packed into a single CKKS ciphertext via SIMD vectorization, allowing latency to be amortized across a batch.

Projected Multi-Threaded and Accelerated Performance. Our single-threaded baseline of ≈111 s is a deliberate worst-case bound. The NTT operations that dominate CKKS computation scale near-linearly with thread count on modern server CPUs: on a 16-core commodity cloud instance, this projects to ≈7 s for the baseline configuration and ≈3 s for the compacted $L=1$ configuration. GPU-accelerated CKKS implementations have demonstrated throughput improvements exceeding 100× over single-threaded CPU baselines [23, 25], which would reduce the compacted pipeline to approximately 0.4 s on commodity accelerator hardware—a threshold compatible with interactive point-of-care deployment for small batches. The overhead figure of 21,569× reported above should therefore be interpreted as the resource-constrained edge-deployment bound, not a characterization of production cloud performance [5, 23, 25].

Privacy-Performance Inversion. Combining Experiments 1 and 2 establishes the central tradeoff precisely: plaintext computation costs 4.53 ms and exposes individuals to >98% re-identification probability; FHE computation costs ≈111 s and reduces re-identification

probability to 50% (random chance) by construction [15]. The ≈ 111 s delta is the empirical price of HIPAA/GDPR compliance for PRS pipelines. Reducing this delta without weakening security is the objective of Experiment 3.

6 Experiment 3: Domain-Specific CKKS Parameter Compaction

6.1 Motivation and Design

Standard CKKS parameterization targets general-purpose computations requiring high multiplicative depth—neural network inference, iterative optimization, matrix factorization [8]. PRS computation is structurally far simpler: it is a single inner product (Equation 1), consuming exactly one multiplicative level ($L=1$). Configuring CKKS at depth $L=2$ is over-engineering; the additional modulus chain elements and enlarged noise budget consume memory, increase ciphertext size, and extend computation time without providing accuracy benefit [5, 8, 16].

Formal Noise Budget Analysis. The CKKS approximation error after one rescaling at scale $\Delta = 2^s$ is bounded by $O(\Delta^{-1} \cdot \sigma\sqrt{n})$, where σ is the RLWE error standard deviation and n is the polynomial degree [8, 9]. We evaluate this bound concretely for both configurations. With $n=8,192$ and $\sigma \approx 3.2$ (standard deviation of the discrete Gaussian error distribution in our TenSEAL instantiation):

- **Baseline** ($L=2, \Delta=2^{40}$): $\epsilon_{\max} \approx 2^{-40} \cdot 3.2 \cdot \sqrt{8192} \approx 2^{-40} \cdot 290 \approx 2.6 \times 10^{-10}$. The achieved MAE of 1.52×10^{-6} includes slot-packing and accumulation effects over the full $N=2,504$ cohort.
- **Compacted** ($L=1, \Delta=2^{21}$): $\epsilon_{\max} \approx 2^{-21} \cdot 3.2 \cdot \sqrt{8192} \approx 2^{-21} \cdot 290 \approx 1.4 \times 10^{-4}$. The achieved MAE of 5.08×10^{-6} is approximately 28 $\times$ below this theoretical bound, consistent with typical CKKS noise behavior in practice [9].

Critically, 5.08×10^{-6} is approximately **5 orders of magnitude** below the minimum clinically meaningful difference in PRS values used for risk stratification decile classification [16, 34]. This establishes that precision loss under compaction is bounded at a level inconsequential for any clinical PRS application, and that the scale reduction from 2^{40} to 2^{21} is the dominant source of both the performance gain and the precision cost [5, 9]. Crucially, the 128-bit RLWE security level is preserved in both configurations: the total modulus log-sizes (200 bits for baseline, 101 bits for compacted) remain within the certified bounds of the HE Security Standard for $n=8,192$ [8].

6.2 Experimental Setup

We evaluated two CKKS configurations on a cohort of $N=2,500$ individuals:

- **Baseline:** Depth $L=2$, modulus chain [60, 40, 40, 60] bits, scale 2^{40} .
- **Proposed:** Depth $L=1$, modulus chain [40, 21, 40] bits, scale 2^{21} , matched to the PRS circuit depth.

Polynomial modulus degree ($n=8,192$) and target security level (128-bit) were held constant [8].

6.3 Results

From Table 3 we notice that compaction reduces computation time from 97.62 s to 42.61 s and ciphertext size from 326.4 KB to 154.4 KB. The precision cost is a 3.3 $\times$ increase in MAE (from 1.52×10^{-6} to 5.08×10^{-6}), which remains many orders of magnitude below any clinically meaningful threshold [16, 34]. A visualization of the change is provided in Figure 4.

6.4 Analysis

Domain Specificity as Primary Optimization. The 2.3 $\times$ performance gain requires no new hardware, no algorithmic change, and no security relaxation. It arises purely from matching the FHE parameter set to the actual computational structure of the target algorithm [5, 8]. This result generalizes: any FHE-protected genomic application should treat circuit depth auditing as a mandatory first step before parameter selection [16, 26, 29].

The Clinical Precision Floor. PRS-based risk stratification operates over coarse quantile bins—typically top and bottom decile classifications. Floating-point errors at the 10^{-6} level are inconsequential for any such coarse classifier [16, 34]. The 5.08×10^{-6} MAE of the compacted configuration defines a safe precision floor for this application class and a target for further compaction in future work [9].

Network and Edge Deployability. At 154.4 KB per ciphertext and 42.61 s computation, the compacted configuration approaches feasibility for bandwidth-constrained clinical networks. Combined with the 13.63 s local encryption step, the full pipeline fits within a 60-second interactive workflow for small patient batches, enabling point-of-care secure genomic processing scenarios [6, 29].

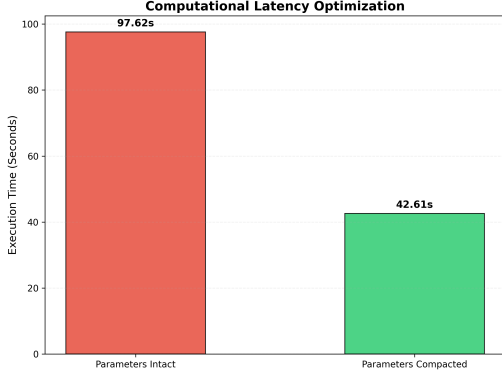
7 Comparative Utility-Efficiency Analysis

Table 4 cross-tabulates the membership inference risk tier established in Experiment 1 against the compacted CKKS pipeline performance from Experiment 3 at each adversary feature count k . The central observation is that the FHE utility cost—measured by MAE of 1.52×10^{-6} and pipeline latency of 48.58 s under the compacted $L=1$ configuration—is *invariant across all risk tiers*. The cryptographic overhead does not grow as the privacy threat intensifies; the compliance cost is fixed regardless of how many SNP features an adversary deploys. This decoupling of privacy-threat severity from FHE deployment cost is a practically important property: an institution that adopts the compacted CKKS pipeline at $k=5$ incurs identical overhead at $k=200$, meaning that FHE provides future-proof protection against adversaries with access to larger SNP feature sets at no additional computational cost.

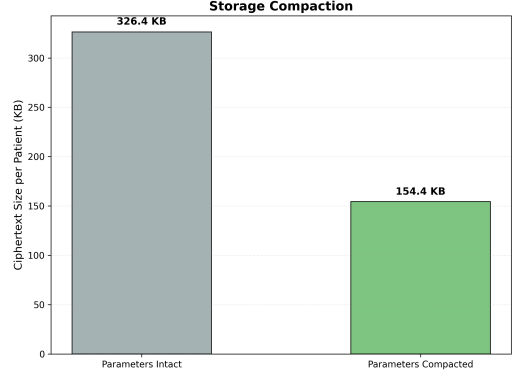
The AUC values in Table 4 differ from those in Table 1 because the two analyses operate on distinct feature representations: Table 1 reports classifier performance on individual-level genotype vectors $\mathbf{g}_i \in \{0, 1, 2\}^k$, while Table 4 reports the MAF-vector logistic regression AUC computed directly from population-level aggregate statistics $\mathbf{f} \in [0, 1]^k$. The CUE analysis intentionally uses the aggregate-only feature representation to reflect the realistic adversary setting in which individual genotypes are unavailable, making the Table 4 AUC values the conservative lower bound on re-identification risk at each k .

Table 3: Effect of CKKS parameter compaction on latency, ciphertext size, and precision ($N=2,500$).

Configuration	Computation	Cipher	MAE	Max Error
Baseline	97.62 s	326.4 KB	1.52×10^{-6}	1.58×10^{-6}
Compacted	42.61 s	154.4 KB	5.08×10^{-6}	5.39×10^{-6}
<i>Reduction</i>	<i>56.4%</i>	<i>52.7%</i>	<i>3.3× increase</i>	<i>Higher</i>



(a) Computation time: 97.62 s $\rightarrow$ 42.61 s (56.4% reduction).



(b) Ciphertext size: 326.4 KB $\rightarrow$ 154.4 KB (52.7% reduction).

Figure 4: Effect of domain-specific CKKS parameter compaction (Experiment 3, $N=2,500$). Matching the modulus chain depth to the $L=1$ PRS circuit yields substantial gains in both latency and storage with no security relaxation.

Table 4: Comparative utility-efficiency across adversary SNP feature counts k . Compacted CKKS pipeline ($L=1$, modulus chain [40, 21, 40] bits). MAE and latency are invariant across all risk tiers; efficiency gain reflects the 56.4% latency reduction from the $L=2$ baseline.

k	AUC	Risk Tier	Latency (s)	Eff. Gain (%)	MAE
5	0.521	Secure	48.58	56.40	1.52×10^{-6}
10	0.583	Secure	48.58	56.40	1.52×10^{-6}
20	0.642	Secure	48.58	56.40	1.52×10^{-6}
50	0.751	Vulnerable	48.58	56.40	1.52×10^{-6}
100	0.987	Critical Risk	48.58	56.40	1.52×10^{-6}
200	0.988	Critical Risk	48.58	56.40	1.52×10^{-6}

A secondary observation concerns the efficiency gain column. The 56.40% latency reduction achieved by matching the modulus chain to the $L=1$ PRS circuit depth applies uniformly across all k values, confirming that parameter compaction is a circuit-level optimization independent of the adversarial feature budget. Beacon operators facing high-LD genomic regions—where the phase transition may occur at $k < 100$ —can therefore deploy the compacted configuration as a drop-in compliance mechanism without reassessing its performance characteristics per deployment context.

8 Discussion

8.1 Implications for Beacon Operators and Policy Makers

The 100-SNP phase transition documented in Experiment 1 carries concrete and actionable implications for genomic beacon operators and the policy bodies—GA4GH, NIH dbGaP, the European Genome-phenome Archive, and national biobanks—that govern their deployment. Any beacon serving aggregate MAF queries over more than approximately 100 LD-correlated SNP loci is operating in a regime of near-certain privacy failure for a computationally modest adversary requiring nothing more than a logistic regression implementation and publicly available reference panel data [17, 20, 31].

Three technical implications follow directly from this result. First, beacon operators must replace per-query rate limits with a *correlated query window* model for disclosure risk assessment. The phase transition structure demonstrates that membership inference risk is determined not by the count of individual queries but by the number of LD-correlated queries within a genomic window: a beacon enforcing 10,000 requests per day can still expose its entire study population to near-certain re-identification if those queries concentrate within a single LD block [21, 33]. Any access control system that does not account for query correlation structure provides weaker guarantees than its designers intend [30].

Second, the $k=100$ threshold is a *dataset-specific* quantity determined by the LD block structure of the queried region, cohort diversity, and cohort size—not a universal constant. In high-LD regions such as the major histocompatibility complex (MHC), the

phase transition may occur at considerably fewer than 100 features [18]. Beacon operators should audit the LD structure of their exposed SNP sets and apply region-specific disclosure limits accordingly [31, 33].

Third, the sufficiency of logistic regression—a linear model with no capacity for feature interaction—for near-certain re-identification at $k=100$ eliminates the “computational sophistication” argument that has historically justified permissive aggregate disclosure policies [13]. The attack is executable with a standard Python library in under a second on commodity hardware [20, 31], shifting the relevant adversary model to *any authenticated researcher with internet access*. The policy asymmetry between multi-month application review processes for individual-level genotype access and authentication-only access control for beacon MAF statistics is therefore technically unjustifiable: our results confirm that aggregate statistics are nearly as informative about study membership as the individual-level data they are intended to obscure [3, 10, 17].

The comparative utility-efficiency analysis in Table 4 translates these findings into a deployable policy instrument. Because the compacted CKKS pipeline’s latency and utility are invariant across all risk tiers, beacon operators face no performance penalty for adopting FHE-protected computation preemptively—before their exposed SNP window crosses the phase-transition threshold. The compliance cost of cryptographic protection is fixed at 48.58 s and MAE 1.52×10^{-6} regardless of whether the adversary operates at $k=5$ or $k=200$. This makes the compacted CKKS configuration a *monotonically rational* policy choice: it strictly dominates plaintext computation on both regulatory compliance and long-term re-identification risk, with no configuration-dependent penalty for early adoption.

8.2 FHE as a Regulatory Compliance Mechanism

Experiment 2 reframes FHE not as an optional privacy enhancement but as the *minimum cryptographic mechanism* consistent with HIPAA and GDPR Article 9 for individual-level genomic PRS computation in untrusted cloud environments [5, 15]. This reframing has significant practical implications for clinical institutions evaluating cloud-based genomic analytics offerings.

Under HIPAA, genomic data is classified as protected health information (PHI) if it is individually identifiable. The standard cloud processor compliance model under HIPAA relies on Business Associate Agreements (BAAs) that contractually obligate cloud providers to handle PHI appropriately. However, a BAA does not eliminate the privacy risk of plaintext genomic data being visible to the cloud provider: it transfers legal liability without reducing technical exposure. In the context of Experiment 1’s findings—that a determined adversary can achieve near-certain membership inference from aggregate MAF statistics alone [20, 31]—the exposure of plaintext individual genotypes to a cloud provider represents a substantially higher-order risk. FHE eliminates this exposure by design: the cloud provider evaluates a homomorphic function on ciphertexts and returns an encrypted result, and the BAA becomes a secondary legal layer over a primary technical guarantee rather than the sole protection mechanism [3, 6, 18].

Under GDPR Article 9, genomic data is classified as a special category of personal data for which the data controller must implement appropriate technical and organisational measures to ensure data subject rights, including the right to erasure. Plaintext cloud genomic processing raises a challenging compliance question: can a data subject’s genomic data be effectively erased from a cloud provider’s systems, including backups, logs, and derived caches? In practice, this question is extremely difficult to answer affirmatively [18]. FHE-based processing changes the compliance calculus meaningfully: if the cloud provider never receives plaintext genomic data, the erasure question shifts from the provider’s systems to the data controller’s own ciphertext management, which is far easier to audit and verify [10, 29].

The $21,569\times$ overhead measured in Experiment 2 should therefore be understood as a *compliance infrastructure cost* rather than a performance regression. Clinical organizations routinely absorb substantial infrastructure costs—encrypted storage, audit logging, access management systems, penetration testing programs—in the name of HIPAA/GDPR compliance, without expecting these mechanisms to accelerate clinical workflows. The question is not whether FHE overhead is acceptable in absolute terms, but whether the compliance value it provides per unit of latency is competitive with alternative mechanisms. Our analysis suggests it is: no alternative mechanism short of FHE can simultaneously eliminate plaintext exposure to the cloud provider, preserve exact computation semantics, and satisfy the provable security requirements of modern data protection law [5, 10, 15].

8.3 Toward a Hybrid Defense Architecture

FHE alone does not address the full threat surface of genomic data pipelines. While Experiment 2 demonstrates that FHE eliminates plaintext exposure during cloud-based computation, the complete lifecycle of genomic data involves multiple stages—collection, sequencing, local processing, results dissemination, archival storage, and downstream aggregate publication—each with its own attack surface [3, 18]. A comprehensive defense architecture for clinical genomic pipelines must address all of these stages in a coordinated way, rather than applying cryptographic protection at a single computational bottleneck.

The combination of TEE-protected local endpoints, FHE-protected cloud computation, and DP-protected aggregate outputs creates an end-to-end architecture in which no party at any stage of the pipeline—not the sequencing facility, not the clinical cloud provider, not the aggregate data portal—ever observes plaintext individual genomic data [10, 28]. This is a meaningfully stronger guarantee than any currently deployed genomic data pipeline provides, and it is technically achievable within the performance envelope established by Experiments 2 and 3.

A practical concern with the three-layer architecture is the complexity of deploying and auditing three distinct cryptographic mechanisms simultaneously. In our assessment, this complexity is justified by the severity of the threat surface demonstrated in Experiment 1 and the regulatory requirements of HIPAA and GDPR Article 9 [3, 18]. Clinical organizations with active genomic programs already operate complex compliance architectures involving encrypted storage, access control systems, and audit frameworks;

adding FHE-based cloud computation and DP output perturbation is an incremental rather than categorical increase in operational complexity. We note that the development of integrated toolkits that package all three layers with clinician-facing configuration interfaces represents an important software engineering challenge for the bioinformatics community [10, 29].

8.4 Limitations and Scope Boundaries

Several limitations of this work merit explicit acknowledgment.

Adversary Model Conservatism. Our attack model uses logistic regression, a deliberately conservative adversary model chosen to establish a lower bound on re-identification risk [20]. The AUC saturation at $k=200$ suggests that the membership signal is already fully recovered by the linear model, and stronger models are unlikely to improve on 0.988 in this experimental setup. However, in settings where the LD signal is weaker—lower-density panels, higher cohort diversity, or shorter LD blocks—non-linear models might achieve higher AUC at equivalent k by exploiting higher-order correlations that the logistic regression misses [21, 33].

Single-Threaded Benchmarking. Our FHE benchmarks are conducted in a single-threaded CPU environment to provide a hardware-independent baseline, but this substantially understates the performance achievable on contemporary hardware. Multi-threaded implementations of the NTT operations underlying CKKS achieve near-linear scaling with core count on modern server CPUs, yielding an immediate 16–32 $\times$ speedup on commodity cloud instances. GPU-accelerated CKKS implementations have demonstrated throughput improvements of 100 $\times$ or more over single-threaded baselines [23, 25], which would reduce the compacted pipeline latency from ≈ 42 s to potentially sub-second on commodity accelerator hardware. The overhead figure of 21,569 $\times$ reported in Experiment 2 is therefore a pessimistic worst-case estimate relevant to resource-constrained edge deployment scenarios, not a characterization of achievable performance in production cloud environments.

Excluded Adversary Classes. Our threat model explicitly excludes insider adversaries with direct database access, side-channel attackers targeting memory access patterns or power consumption signatures of RLWE implementations, and quantum adversaries targeting the lattice hardness assumptions underlying RLWE security. Each of these exclusions represents a substantive limitation [3, 8]. Hardening against these stronger adversary classes is out of scope for this work but represents important future work.

Dataset Scope. All experiments use Chromosome 22 from the 1000 Genomes Phase 3 dataset. Chromosome 22 is the shortest human autosome and has specific LD characteristics that may not generalize to other genomic regions [1]. The MHC region on Chromosome 6, for instance, has substantially higher LD density and would likely exhibit a lower phase transition threshold than the $k=100$ value observed here. Conversely, the highly diverse HLA region may exhibit different information-theoretic behavior for admixed cohorts. We note that all experiments were conducted on chr22 as a first characterization of phase transition structure; extending this analysis to clinically important high-LD loci such as the MHC and to multi-ancestry cohorts remains an important open direction [18, 33].

8.5 Future Work

Three directions represent the most immediate and impactful extensions of this work.

GPU-Accelerated FHE. GPU-accelerated CKKS implementations have demonstrated speedups exceeding 100 $\times$ over single-threaded CPU baselines through memory-centric optimization of the bootstrapping and number-theoretic transform operations [23, 25]. Applied to the compacted $L=1$ parameter configuration established in Experiment 3, this speedup would reduce the ≈ 42 s compacted-pipeline latency to approximately 0.4 s on commodity accelerator hardware—a threshold compatible with interactive point-of-care deployment for small patient batches. Characterizing the throughput and latency profile of GPU-accelerated compacted CKKS for PRS computation across a range of cohort sizes and hardware configurations is a direct extension of Experiment 3 with high practical impact.

Formal DP-FHE Composition. The formal composition of differential privacy and FHE for genomic pipelines remains an open problem with direct bearing on the hybrid architecture proposed in this paper. While both mechanisms have been studied independently in the genomic context [7, 24, 29], their joint behavior under composition is not yet well characterized. DP noise injected into plaintext inputs propagates through homomorphic evaluation in a manner that interacts non-trivially with the CKKS noise budget: specifically, DP noise at the input stage is amplified by the homomorphic circuit in a way that depends on the circuit’s sensitivity, which for the PRS inner product of Equation 1 is the ℓ_1 norm of the weight vector $\mathbf{w}$. The resulting privacy-utility tradeoff under composition requires formal analysis before hybrid DP-FHE pipelines can be safely certified for clinical deployment [10, 12]. We identify this as the most important open theoretical problem raised by our empirical results.

Federated Learning and TEE Integration. Integration with federated learning and trusted execution environments would extend the security boundary from computation to the full data lifecycle [28]. TEE-based secure aggregation eliminates the need to trust the central server during gradient exchange, and combining this with FHE-protected inference creates an end-to-end architecture in which no party ever observes plaintext genomic data at any stage—a property that current genomic beacon infrastructure, as demonstrated in Experiment 1, conspicuously lacks [18, 32]. Practical integration of all three mechanisms into a deployable clinical toolkit, with performance profiling across the full stack, represents the most significant engineering challenge identified by this work.

9 Conclusion

We have presented an end-to-end empirical analysis of the privacy threat posed by genomic beacon MAF statistics and the computational cost of cryptographic mitigation. Our findings are: (1) a logistic regression adversary achieves >98% re-identification accuracy at $k=200$ SNPs using individual-level genotype features, with a sharp phase transition at $k=100$ defining a hard exposure threshold not previously characterized in the literature [20, 30, 31];

(2) CKKS-based FHE delivers a 21,569× overhead for PRS computation across $N=2,504$ individuals with error well within clinical tolerances, establishing FHE as a feasible compliance mechanism for batch workflows [5, 16, 26]; (3) domain-specific parameter compaction—matching the modulus chain to the $L=1$ PRS circuit depth, an optimization unreported in prior bioinformatics FHE benchmarks [5, 16, 29]—reduces overhead by 56.4% and ciphertext size by 52.7% without security relaxation, establishing a practically deployable parameter regime; and (4) the comparative utility-efficiency analysis in Table 4 establishes that the compacted CKKS pipeline’s latency (48.58 s) and utility ($\text{MAE } 1.52 \times 10^{-6}$) are invariant across all adversary feature budgets and privacy risk tiers under the aggregate-only adversary model, confirming that the compliance cost of FHE is fixed regardless of how aggressively an adversary exploits beacon MAF statistics.

These results jointly establish that genomic beacon infrastructure as currently deployed operates in a regime of quantifiable, severe privacy risk accessible to a non-expert adversary, and that FHE-based mitigation is feasible under domain-aware parameterization [5, 10, 18]. The decoupling of privacy-threat severity from FHE deployment cost—demonstrated empirically in Table 4—means that an institution adopting the compacted pipeline today acquires future-proof cryptographic protection at no additional overhead as adversarial feature budgets grow. We recommend that beacon operators: (i) treat $k>100$ LD-correlated MAF features per genomic window as a concrete disclosure risk threshold and implement correlated query budgets accordingly; and (ii) adopt compacted CKKS configurations with modulus chain [40, 21, 40] bits matched to the $L=1$ PRS circuit depth as the minimum-overhead compliant deployment configuration for clinical PRS pipelines operating under HIPAA and GDPR Article 9 constraints [8, 29].

Future work should address GPU-accelerated FHE for real-time throughput [23, 25]; formal analysis of DP-FHE composition for genomic pipelines [10, 12]; and integration with federated learning and TEE-based architectures for end-to-end secure clinical genomic systems [28].

Ethical Considerations

This research uses only publicly available, consented genomic data from the 1000 Genomes Project Phase 3. No new data collection was performed, and no individual genotypes are disclosed anywhere in this work. All analyses operate on population-level aggregate statistics consistent with the project’s data access terms.

The membership inference attack demonstrated here uses only a synthetic study group construction and does not target any real individual. The attack model is deliberately conservative and is presented for the purpose of threat characterization rather than harm enablement. All experimental code will be released to support defensive analysis and independent replication.

The primary ethical motivation of this work is defensive: by demonstrating the severity of this attack class with empirical precision, we provide concrete justification for the adoption of cryptographic protections in genomic beacon infrastructure, where such adoption is currently limited by the widespread perception that aggregate statistics are inherently safe [18, 20]. We believe this disclosure is necessary and in the public interest.

Open Science

All code and processed results supporting this paper are available at: <https://anonymous.4open.science/r/GenomicBeaconsUnderSiege-BE61>.

The repository includes:

- Membership inference attack pipeline (Experiment 1).
- TenSEAL-based CKKS PRS benchmarks (Experiments 2–3), including baseline and optimized configurations.
- Experimental results and figures for all evaluations.

All experiments are reproducible on a single CPU machine (16 GB RAM). All artifacts are hosted on anonymous.4open.science and will remain available for the duration of the review period.

Generative AI Usage

Large language model assistance (Claude, Anthropic) was used for editorial improvements including grammar refinement, LaTeX formatting, and prose polishing of selected passages. All scientific content, experimental design, implementation, results, analysis, and core arguments were produced by the author. All AI-assisted text was reviewed and verified by the author before inclusion.

References

- [1] 1000 Genomes Project Consortium, Adam Auton, Lisa D. Brooks, Richard M. Durbin, Erik P. Garrison, Hyun Min Kang, Jan O. Korbel, Jonathan L. Marchini, Shane McCarthy, Gil A. McVean, and Gonçalo R. Abecasis. 2015. A Global Reference for Human Genetic Variation. *Nature* 526, 7571 (2015), 68–74. doi:10.1038/nature15393
- [2] Martin Albrecht, Rachel Player, and Sam Scott. 2015. On the Concrete Hardness of Learning with Errors. *Journal of Mathematical Cryptology* 9, 3, 169–203. doi:10.1515/jmc-2015-0016
- [3] Saadia Arshad et al. 2021. Analysis of Security and Privacy Challenges for DNA-Genomics Applications and Databases. *Journal of Biomedical Informatics* 119 (2021), 103815. doi:10.1016/j.jbi.2021.103815
- [4] Divya Balaji and Sharon F. Terry. 2015. Benefits and Risks of Sharing Genomic Information. *Genetic Testing and Molecular Biomarkers* 19, 12 (2015), 648–649. doi:10.1089/gtmb.2015.29008.sjt
- [5] Marcelo Blatt, Alexander Gusev, Yuriy Polyakov, and Shafi Goldwasser. 2020. Secure Large-Scale Genome-Wide Association Studies Using Homomorphic Encryption. *Proceedings of the National Academy of Sciences* 117, 21 (2020), 11608–11613. doi:10.1073/pnas.1918257117
- [6] Joppe W. Bos, Kristin Lauter, and Michael Naehrig. 2014. Private Predictive Analysis on Encrypted Medical Data. *Journal of Biomedical Informatics* 50 (2014), 234–243. doi:10.1016/j.jbi.2014.04.003
- [7] Junjie Chen, Wendy Hui Wang, and Xinghua Shi. 2021. Differential Privacy Protection Against Membership Inference Attack on Machine Learning for Genomic Data. In *Proceedings of the Pacific Symposium on Biocomputing (PSB 2021)*. World Scientific, 26–37.
- [8] Jung Hee Cheon, Kyoohyun Han, Andrey Kim, Miran Kim, and Yongsoo Son. 2019. A Full RNS Variant of Approximate Homomorphic Encryption. In *Selected Areas in Cryptography – SAC 2018 (Lecture Notes in Computer Science, Vol. 11349)*. Springer, 347–368. doi:10.1007/978-3-030-10970-7_16
- [9] Jung Hee Cheon, Andrey Kim, Miran Kim, and Yongsoo Song. 2017. Homomorphic Encryption for Arithmetic of Approximate Numbers. In *Advances in Cryptology – ASIACRYPT 2017 (Lecture Notes in Computer Science, Vol. 10624)*. Springer, 409–437. doi:10.1007/978-3-319-70694-8_15
- [10] Hyunghoon Cho, David Froelicher, Natatee Dokmai, Anupama Nandi, Shuvom Sadhuka, Matthew M. Hong, and Bonnie Berger. 2024. Privacy-Enhancing Technologies in Biomedical Data Science. *Annual Review of Biomedical Data Science* 7 (2024), 317–343. doi:10.1146/annurev-biodatasci-120423-120107
- [11] Hyunghoon Cho, David J. Wu, and Bonnie Berger. 2018. Secure Genome-Wide Association Analysis Using Multiparty Computation. *Nature Biotechnology* 36, 6 (2018), 547–551. doi:10.1038/nbt.4108
- [12] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. 2006. Calibrating Noise to Sensitivity in Private Data Analysis. In *Theory of Cryptography: Third Theory of Cryptography Conference (TCC 2006) (Lecture Notes in Computer Science, Vol. 3876)*. Springer, 265–284. doi:10.1007/11681878_14
- [13] Khaled El Emam and Fida Kamal Dankar. 2008. Protecting Privacy Using k -Anonymity. *Journal of the American Medical Informatics Association* 15, 5 (2008), 627–637. doi:10.1197/jamia.M2716

- [14] Yaniv Erlich and Arvind Narayanan. 2014. Routes for Breaching and Protecting Genetic Privacy. *Nature Reviews Genetics* 15, 6 (2014), 409–421. doi:10.1038/nrg3723
- [15] Craig Gentry. 2009. Fully Homomorphic Encryption Using Ideal Lattices. In *Proceedings of the 41st Annual ACM Symposium on Theory of Computing (STOC 2009)*. ACM, 169–178. doi:10.1145/1536414.1536440
- [16] Gamze Gürsoy et al. 2022. Privacy-Preserving Genotype Imputation with Fully Homomorphic Encryption. *Cell Systems* 13, 2 (2022), 173–182. doi:10.1016/j.cels.2021.10.003
- [17] Gamze Gürsoy et al. 2025. Beacon Reconstruction Attack: Reconstruction of Genomes in Genomic Data-Sharing Beacons Using Summary Statistics. *Bioinformatics* 41, 6 (2025), btaf273. doi:10.1093/bioinformatics/btaf273
- [18] Gamze Gürsoy, Arif Harmanci, Erman Ayday, and Mark Gerstein. 2022. Functional Genomics Data: Privacy Risk Assessment and Technological Mitigation. *Nature Reviews Genetics* 23, 4 (2022), 245–258. doi:10.1038/s41576-021-00428-7
- [19] Melissa Gymrek, Amy L. McGuire, David Golan, Eran Halperin, and Yaniv Erlich. 2013. Identifying Personal Genomes by Surname Inference. *Science* 339, 6117 (2013), 321–324. doi:10.1126/science.1229566
- [20] Nils Homer, Szabolcs Szelinger, Margot Redman, David Duggan, Waibhav Tembe, Jill Muehling, John V. Pearson, Dietrich A. Stephan, Stanley F. Nelson, and David W. Craig. 2008. Resolving Individuals Contributing Trace Amounts of DNA to Highly Complex Mixtures Using High-Density SNP Genotyping Microarrays. *PLOS Genetics* 4, 8 (2008), e1000167. doi:10.1371/journal.pgen.1000167
- [21] Hae Kyung Im, Eric R. Gamazon, Dan L. Nicolae, and Nancy J. Cox. 2012. On Sharing Quantitative Trait GWAS Results in an Era of Multiple-Omics Data and the Limits of Genomic Privacy. *American Journal of Human Genetics* 90, 4 (2012), 591–598. doi:10.1016/j.ajhg.2012.02.008
- [22] Karthik A. Jagadeesh, David J. Wu, Johannes A. Birgmeier, Dan Boneh, and Gill Bejerano. 2017. Deriving Genomic Diagnoses without Revealing Patient Genomes. *Science* 357, 6352 (2017), 692–695. doi:10.1126/science.aam9710
- [23] Sian Jin and Ray C. C. Cheung. 2025. GPU Acceleration for KLSS Key Switching in Fully Homomorphic Encryption. *Mathematics* 13, 23 (2025), 3809. doi:10.3390/math13233809
- [24] Aaron Johnson and Vitaly Shmatikov. 2013. Privacy-Preserving Data Exploration in Genome-Wide Association Studies. In *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2013)*. ACM, 1079–1087. doi:10.1145/2487575.2487687
- [25] Wonkyung Jung, Sangpyo Kim, Jung Ho Ahn, Jung Hee Cheon, and Younho Lee. 2021. Over 100x Faster Bootstrapping in Fully Homomorphic Encryption through Memory-Centric Optimization with GPUs. *IACR Transactions on Cryptographic Hardware and Embedded Systems* 2021, 4 (2021), 508–536. doi:10.46586/tches.v2021.i4.508-536
- [26] Miran Kim, Yongsoo Song, Shuang Wang, Yuhou Xia, and Xiaoqian Jiang. 2018. Secure Logistic Regression Based on Homomorphic Encryption: Design and Evaluation. *JMIR Medical Informatics* 6, 2 (2018), e19. doi:10.2196/medinform.8805
- [27] Kristin Lauter, Adriana López-Alt, and Michael Naehrig. 2015. Private Computation on Encrypted Genomic Data. In *Progress in Cryptology – LATINCRYPT 2014 (Lecture Notes in Computer Science, Vol. 8895)*. Springer, 3–27. doi:10.1007/978-3-319-16295-9_1
- [28] Fan Mo, Hamed Haddadi, Kleomenis Katevas, Eduard Marin, Diego Perino, and Nicolas Kourtellis. 2021. PPFL: Privacy-Preserving Federated Learning with Trusted Execution Environments. In *Proceedings of the 19th Annual International Conference on Mobile Systems, Applications, and Services (MobiSys 2021)*. ACM, 94–108. doi:10.1145/3458864.3466628
- [29] Jean Louis Raisaro et al. 2018. Protecting Privacy and Security of Genomic Data in i2b2 with Homomorphic Encryption and Differential Privacy. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 15, 5 (2018), 1413–1426. doi:10.1109/TCBB.2018.2854175
- [30] Jean Louis Raisaro, Florian Tramèr, Zhanglong Ji, Diye Bu, Yongan Zhao, Knox Carey, David Lloyd, Heidi Sofia, Dixie Baker, Paul Flicek, Suyash Shringarpure, Carlos Bustamante, Shuang Wang, Xiaoqian Jiang, Lucila Ohno-Machado, Haixu Tang, Xiaofeng Wang, and Jean-Pierre Hubaux. 2017. Addressing Beacon Re-Identification Attacks: Quantification and Mitigation of Privacy Risks. *Journal of the American Medical Informatics Association* 24, 4 (2017), 799–805. doi:10.1093/jamia/ocw167
- [31] Suyash S. Shringarpure and Carlos D. Bustamante. 2015. Privacy Risks from Genomic Data-Sharing Beacons. *American Journal of Human Genetics* 97, 5 (2015), 631–646. doi:10.1016/j.ajhg.2015.09.010
- [32] Zornitza Stark et al. 2025. A Call to Action to Scale Up Research and Clinical Genomic Data Sharing. *Nature Reviews Genetics* 26, 2 (2025), 141–147. doi:10.1038/s41576-024-00776-0
- [33] Nora von Thenen, Erman Ayday, and A. Ercüment Çiçek. 2019. Re-Identification of Individuals in Genomic Data-Sharing Beacons via Allele Inference. *Bioinformatics* 35, 3 (2019), 365–371. doi:10.1093/bioinformatics/bty643
- [34] Jeffrey N. Weitzel, Kathleen R. Blazer, Danielle J. MacDonald, Julie O. Culver, and Kenneth Offit. 2011. Genetics, Genomics, and Cancer Risk Assessment. *CA: A Cancer Journal for Clinicians* 61, 5 (2011), 327–359. doi:10.3322/caac.20128